

Nicotine Dose Curve

A concise methods companion to the interactive smoker-versus-vaper 24-hour comparison

Purpose

This report documents a deliberately simple teaching model for comparing two delivery rhythms: 20 cigarette-like absorbed-dose boluses between 07:00 and 23:00, and 120 smaller vape-like top-ups across the same window. The baseline gives each pattern 24 mg of nicotine absorbed into systemic circulation over the day. It is an illustrative pharmacokinetic model, not a clinical prediction or a dose recommendation.

Model

Nicotine is represented by a one-compartment pool with first-order elimination. After each absorbed dose D at time t_i , its contribution is $D \exp[-k(t-t_i)]$ for $t \geq t_i$, where $k = \ln(2)/t_{1/2}$. The plotted signal is the sum of all active contributions: $C(t) = \sum D_i e^{-kt-t_i}$. Baseline half-life is 2 h, consistent with commonly cited plasma nicotine estimates of roughly 2–3 h.

Baseline assumptions

Parameter	Smoker pattern	Vape pattern
Daily absorbed dose	24 mg	24 mg
Events	20 boluses	120 micro-doses
Window	07:00–23:00	07:00–23:00
Dose per event	1.20 mg	0.20 mg
Half-life	2.0 h	2.0 h

Baseline results

Metric	20 boluses	120 micro-doses
Peak model signal	4.73	4.38
Lowest sampled signal	0.30	0.19
24 h AUC (model units·h)	68.4	68.5
07:00 next-day residual	0.30	0.27

The exact values depend on the event timing convention and numerical sampling. The interpretation is more stable than the last decimal: repeated small inputs smooth peak-to-trough variability, while equal absorbed daily dose produces similar exposure burden under the same elimination assumption.

Interpretation

Product nicotine is not absorbed nicotine. A cigarette can contain several milligrams of nicotine in tobacco while systemic absorption is commonly nearer 1–2 mg, with substantial person-to-person variation. For e-liquid, mg/mL describes reservoir content; aerosol generation, inhalation, deposition, exhalation, and device behavior determine what reaches circulation.

Bolus versus quasi-continuous dosing. Cigarette sessions deliver a rapid pulse, so concentrations rise and fall repeatedly. Frequent vape top-ups approximate a smoother input function. Smoother exposure can reduce visible troughs without reducing total absorbed dose.

Self-titration matters. Experienced users can change puff duration, intensity, and interval. Reviews of e-cigarette pharmacology report that user experience and behavior strongly influence systemic nicotine, so a fixed ratio between products should be treated as a scenario control, not a universal conversion.

Overnight decay. After 23:00, no new doses are added. With a 2 h half-life, the model halves repeatedly overnight; a residual remains at 07:00 because elimination is continuous rather than instantaneous.

Limitations

This model omits absorption delays, lung and buccal compartments, distribution, metabolites, tolerance, withdrawal, product formulation, device power, puff topography, body size, metabolism, and uncertainty in the daily dose. It does not predict blood nicotine, impairment, cardiovascular response, dependence, or health risk. The vape-to-smoker ratio is a user-controlled scenario parameter. AUC here is the area under an arbitrary model signal, not a measured clinical exposure metric.

Sources and provenance

The interactive baseline follows the prior Wolfram-derived setup supplied in the referenced conversation. General pharmacology context: NCBI Bookshelf, *Clearing the Smoke: Nicotine Pharmacology*, which describes plasma nicotine half-life around 2–3 hours and daytime accumulation; and Dawkins et al., *Electronic cigarettes: clinical pharmacology, Pharmacology & Therapeutics* (2014), a review emphasizing device and user-experience effects on systemic delivery. Links: ncbi.nlm.nih.gov/books/NBK222359 and pmc.ncbi.nlm.nih.gov/articles/PMC3995273.

Use the interactive companion to change daily dose, event counts, the vape-to-smoker ratio, and half-life. Treat the output as a visual reasoning aid: it clarifies timing and accumulation, while real-world exposure requires measurement and clinical context.

ニコチン投与曲線

紙巻きたばこ電子たばこの24時間モデル：方法と解釈

目的

これは学習用の薬物動態モデルです。07:00から23:00まで、20回の紙巻きたばこ型ボース投与と、120回の電子たばこ型の微量追加投与を比較します。基準条件では、どちらも体循環に吸収されるニコチンを1日24 mgとします。臨床予測や用量の推奨ではありません。

モデル

ニコチンを1コンパートメントのプールとして表し、一次消失を仮定します。半減期を2時間、消失速度を $k = \ln(2) / \text{半減期}$ とし、各投与 D の寄与を $D \exp[-k(t-t_i)]$ とします。曲線は、時刻 t までに投与されたすべての寄与の合計です。

基準条件

紙巻きたばこ：20回、1回1.20 mg、07:00 – 23:00。

電子たばこ：120回、1回0.20 mg、07:00 – 23:00。

半減期：2.0時間。日吸収量：24 mg。

基準結果

モデル信号のピーク：紙巻きたばこ 約4.73、電子たばこ 約4.38。

24時間AUC：紙巻きたばこ 約68.4、電子たばこ 約68.5（モデル単位・時）。

翌朝07:00の残留：紙巻きたばこ 約0.30、電子たばこ 約0.27。

解釈と限界

ボース投与はピークと谷を大きくし、頻回の微量投与は曲線を滑らかにします。吸収量が同じなら、滑らかさは総曝露が少ないことを意味しません。製品中のニコチン量、リキッドの mg/mL、エアロゾル収率、吸収された量は別の量です。実際の曝露は吸入方法、機器、製剤、吸収、代謝、耐性、自己調整などに左右されます。このモデルは血中濃度、依存、心血管反応、健康リスクを予測しません。

出典

基準設定は参照された会話の計算条件を引き継いでいます。背景資料：NCBI Bookshelf「Nicotine Pharmacology」および Dawkins らによる電子たばこの臨床薬理レビュー（2014）。